1)

a) False (see p. 4 in the lecture notes)

b) True (see p. 11 in the lecture notes)

c) True (see Theorem 3.10 b) in the lecture notes)

2)

a) There exist $\mu_A, \mu_B, \mu_C \in \mathbb{R}$ and $\sigma \in (0, \infty)$ such that the age of a player from team $j \in \{A, B, C\}$ is $\text{Normal}(\mu_j, \sigma^2)$-distributed. The ages of different players are independent.

b) For $j \in \{A, B, C\}$ let $\hat{\mu}_j$ be the sample mean of team j. We have

$$\hat{\mu}_A = 26.9965$$

$$\hat{\mu}_B = 26.9965 + 2.5845 = 29.5810$$

$$\hat{\mu}_C = 26.9965 - 0.9060 = 26.0905.$$

The estimate $\hat{\sigma}$ for σ is

$$\hat{\sigma} = 3.447.$$

c) We test

$$H_0: \mu_A = \mu_B = \mu_C$$

against

$$H_1: \mu_i \neq \mu_j \text{ for some } i, j \in \{A, B, C\}.$$

Since the p-value 0.001812 is less than 0.05, we can reject H_0 at level 0.05, i.e. there is a difference between the teams.

d) We test

$H_0^{(1)}: \mu_A = \mu_B$ against $H_1^{(1)}: \mu_A \neq \mu_B$,

$H_0^{(2)}: \mu_A = \mu_C$ against $H_1^{(2)}: \mu_A \neq \mu_C$,

$H_0^{(3)}: \mu_B = \mu_C$ against $H_1^{(3)}: \mu_B \neq \mu_C$.

Since their p-values are below 0.05, we can reject $H_0^{(1)}$ and $H_0^{(3)}$ at level 0.05, while we cannot reject $H_0^{(2)}$.

We can conclude that there is a difference between teams A and B and teams B and C. In fact, the players of B are older than those of A and C. The relation between A and C is not clear.

3)

a) A least square estimator $(\hat{\alpha}_0, \hat{\alpha}_1)^T$ for $(\alpha_0, \alpha_1)^T$ is a solution of

$$\min_{\tilde{\alpha}_0, \tilde{\alpha}_1 \in \mathbb{R}} \quad (10-\tilde{\alpha}_0)^2 + (7.5-\tilde{\alpha}_0 e^{-\tilde{\alpha}_1})^2 + (5-\tilde{\alpha}_0 e^{-2\tilde{\alpha}_1})^2 + (3-\tilde{\alpha}_0 e^{-3\tilde{\alpha}_1})^2.$$

b) We have that

$$P(\text{the part is destroyed}) = \frac{\exp(\beta_0+\beta_1 t)}{1+\exp(\beta_0+\beta_1 t)}.$$

c) We know that for

$H_0: \langle a, \gamma \rangle = c_0$ against $H_1: \langle a, \gamma \rangle \neq c_0$,

$$\varphi_{\neq}(Y) = \begin{cases} \text{cannot reject}, & c_0 \in I^{\neq}_{a,\hat{\gamma}}, \\ \text{reject}, & c_0 \notin I^{\neq}_{a,\hat{\gamma}}, \end{cases}$$

with

$$I^{\neq}_{a,\hat{\gamma}} = \left(\langle a, \hat{\gamma} \rangle - t_{n-1-k, 1-\frac{\alpha}{2}} \hat{\sigma} \sqrt{a^T (A^T A)^{-1} a}, \quad \langle a, \hat{\gamma} \rangle + t_{n-1-k, 1-\frac{\alpha}{2}} \hat{\sigma} \sqrt{a^T (A^T A)^{-1} a} \right)$$

is a level-α test. We have $k=2$, $a=(0,1,-1)^T$ and $c_0=0$ and obtain

$$a^T (A^T A)^{-1} a = (0,1,-1) \begin{pmatrix} n & & \\ & 2 & \\ & & 2 \end{pmatrix}^{-1} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

$$= (0,1,-1) \begin{pmatrix} \frac{1}{n} & & \\ & \frac{1}{2} & \\ & & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = \frac{1}{2} + \frac{1}{2} = 1.$$

Thus,

$$I^{\neq}_{a,\hat{\gamma}} = \left(\hat{\gamma}_1 - \hat{\gamma}_2 - t_{n-3, 1-\frac{\alpha}{2}} \hat{\sigma}, \ \hat{\gamma}_1 - \hat{\gamma}_2 + t_{n-3, 1-\frac{\alpha}{2}} \hat{\sigma} \right),$$

which completes the proof.